# Andreas Küpfer

**Computational Social Scientist**

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My research focuses on (multimodal) political communication, combining computational with traditional methods to address substantial questions on party competition and political behaviour.

## Current Position

2022-present **Research Associate**
Technical University of Darmstadt
Institute for Political Science (Christian Stecker)

## Education

2022-2026 **Ph.D. Political Science**, University of Mannheim
Supervisors: Christian Stecker and Christian Arnold
Committee: Thomas Gschwend, Marc Ratkovic, Christian Stecker
Thesis: *Multimodal Data and Open Science: Innovations in the Study of Political Communication and Behaviour*

2019-2022 **M.Sc. Data Science**, University of Mannheim
Supervisors: Thomas Gschwend and Denis Cohen
Thesis: *When local context trumps party unity: Using Twitter and NLP to explain geographical heterogeneity in legislators' policy communication*

2015-2019 **B.Sc. Business Information Systems**, University of Applied Sciences Karlsruhe

2017 **ERASMUS Student (Studies Abroad), B.Sc. Informatics**, Metropolia University of Applied Sciences in Helsinki, Finland

## Publications

2025 **The sound of party competition: how applause reflects unity, disagreement, and the electoral cycle in parliaments**
*A. Küpfer*, J. Müller, C. Stecker
West European Politics

2025 **The Politics of Seeking and Avoiding Discourse in Parliament**
E. Koch, *A. Küpfer*
European Journal of Political Research

2024 **Nonrandom Tweet Mortality and Data Access Restrictions: Compromising the Replication of Sensitive Twitter Studies**
*A. Küpfer*
Political Analysis

2024 **The Role of Hyperparameters in Machine Learning Models and How to Tune Them**
C. Arnold, L. Biedebach, *A. Küpfer*, M. Neunhoeffer
Political Science Research and Methods

# Working Papers

**Look Who's Confronting: Opposition Status, Gender, and the Far Right**
A. Küpfer

**Structuring Quantitative Image Analysis with Object Prominence**
with C. Arnold

**Alignment Helps Make the Most of Multimodal Data**
with C. Arnold

**Measuring Media Slant through Image Analysis**
with C. Arnold, O. Rittmann, M. Torres

**A Pipeline for Extracting Data from Videos of Complex Political Events**
with M. Holman, T. Simko

**When Words Divide: MPs as Polarisation Entrepreneurs on the Digital Civic Platform abgeordnetenwatch.de**
with C. Sheldon, C. Arnold

# Research Visits and Summer Schools

01/2026-10/2026 **Centre for Artificial Intelligence in Government (CAIG) (Christian Arnold)**, University of Birmingham, UK

10/2024-10/2025 **Data and Methods Unit**, Mannheim Centre for European Research (MZES), University of Mannheim, Germany

06/2024 **Video Research Institute, Center for C-SPAN Scholarship & Engagement (CCSE) (Bryce Dietrich)**, Purdue University, US

02/2024-03/2024 **Centre for Artificial Intelligence in Government (CAIG) (Slava Jankin)**, University of Birmingham, UK

# Grants, Certificates, and Awards

2025 **Grant: APSA Political Communication Travel Award, 600$**

2025 **Certificate: TU Darmstadt Higher Education Teaching Certificates: Professional Competence Level (200 work units)**

2024 **Grant: German Research Foundation (DFG) Grant "Analysing Legislative Politics with Aligned Text, Audio, and Image Data" with Christian Arnold and Christian Stecker, 285'000€**

2024 **Grant: APSA Political Communication SPARK Travel Award, 750$**

2023 **Award: Athene Award for Excellent Teaching "Promoting data literacy, strengthening critical thinking, taking individual needs into account, integrating modern tools" (TU Darmstadt), 2'000€**

# Teaching and Supervision

**Master Thesis Supervisions: 1**

**Student Assistant Supervisions: 5**

2024 **Machine Learning for Social Scientists**, PhD graduate school course, MZES (with Ruben Bach)

2024 **Introduction to Analysis and Comparison of Political Systems**, Undergraduate proseminar, Technical University of Darmstadt

2023 **Quantitative Text Analysis in R**, Postgraduate seminar, Technical University of Darmstadt

2023 **Data Analysis: From Data Crawling to Visualization**, Guest lecture (180 minutes), National University of Kyiv-Mohyla Academy

2023 **Getting started with Python: A how-to guide for social scientists**, Workshop with two sessions à 90 minutes, MZES Social Science Data Lab (with Ruben Bach)

2022, 2023 **Introduction to Quantitative Methods in R**, Postgraduate seminar, Technical University of Darmstadt

2022 **Collection, Management, and Analysis of Twitter Data: Using the Twitter API for Academic Research and BERT**, Workshop (90 minutes), MZES Social Science Data Lab

2017 **IT Security and Operating Systems**, Undergraduate tutorial, University of Applied Sciences Karlsruhe

2016 **Programming II**, Undergraduate tutorial, University of Applied Sciences Karlsruhe

2016 **Introduction to Business Informatics**, Undergraduate tutorial, University of Applied Sciences Karlsruhe

# Research Dissemination

## Invited Talks

2025 **Data Science Brown Bag Series**, Hertie School, Berlin, Germany

2025 **Social Data Science Seminar**, University of Mannheim, Germany

2024 **Mittelbau Colloquium**, MZES, University of Mannheim, Germany

2024 **Text-as-Data (TADA) Speaker Series**, Virtual

## Conferences

2025 **APSA (Main Conference and Political Communication Pre-Conference)**[p], Vancouver, Canada

2025 **COMPTEXT**[p,c,d], Vienna, Austria

2025 **PSA**, Birmingham, UK

2024 **APSA**[p], Philadelphia, US

2024 **EPSA**[p], Cologne, Germany

2024 **COMPTEXT**[p], Amsterdam, Netherlands

2024 **IASGP Annual Conference**[c], Darmstadt, Germany

2023 **IASGP Annual Conference**[p], Mannheim, Germany

2023 **ECPR Standing Group on Parliaments**[p,d,c], Vienna, Austria

2023 **EPSA**[p,d], Glasgow, UK

2023 **COMPTEXT**[p,c], Glasgow, UK

2022 **DVPW Group Comparative Research on Parliaments**[p], Bremen, Germany

2022 **EPSA**[p], Prague, Czech Republic

2022 **PolMeth Europe**[p], Hamburg, Germany

2021 **PolMeth Europe**, Virtual

**p = Presenter, d = Discussant, c = Panel Chair/Roundtable**

## Press

2025 **Applaus-O-Meter: Zwei Fraktionen applaudieren vor allem sich selbst**, Frankfurter Allgemeine Sonntagszeitung (FAS) (with Jochen Müller and Christian Stecker)

2025 **When do politicians engage in discourse – and when do they avoid it?**, LSE EUROPP – European Politics and Policy Blog (with Elias Koch)

2023 **Der Frankfurter Wahlkompass VOTO zur OB-Wahl 2023**, FAZ (with Thilo Dieing and Christian Stecker)

2023 **Wahlrecht ab 16, mehr autofreie Zonen und gebührenfreie Kinderbetreuung? Die Positionen der Kandidierenden zur Oberbürgermeisterwahl in Darmstadt in der Online-Wahlhilfe VOTO**, Darmstädter Echo (with Christian Stecker)

2023 **Wahlkompass bietet Orientierung im Kandidatendschungel (FAZ)**, (with Christian Stecker)

2023 **Viele Köpfe und ein breites Angebot. Die Positionen der Kandidierenden zur Frankfurter Oberbürgermeisterwahl**, FAZ (with Christian Stecker)

## Web Publications

2023 **BERT and Explainable AI**, Methods Bites, Blog of the Mannheim Centre for European Research (MZES) Social Science Data Lab

2022 **Collection, Management, and Analysis of Twitter Data**, Methods Bites, Blog of the Mannheim Centre for European Research (MZES) Social Science Data Lab

## Software

2022 **triviz**, Triangular visualizations for the significance of pairwise differences from multigroup comparisons (Denis Cohen and *Andreas Küpfer*)

# Referee

**Computational Communication Research • Computational Linguistics for the Political and Social Sciences at KONVENS (Workshop), 2024 & 2025 • Information, Communication and Society • Nature Scientific Data • Political Analysis • Politische Vierteljahresschrift • Zeitschrift für Politikwissenschaft**

# Service

**Co-Organizer/Contributor of the Social Science Data Lab (SSDL) at MZES (since 03/2021) • Co-Organizer of the 2024 ECPR Summer School on Parliaments (08/2024) • Member of the decision committee for the German public-private scholarship (Deutschlandstipendium) (since 11/2023) • Data infrastructure representative (since 11/2023)**

# Selected Work Experience (Academia)

03/2021-09/2024 **Student Research Assistant at Social Science Data Lab**, Mannheim Centre for European Research (MZES) (Denis Cohen)

10/2020-12/2021 **Student Research Assistant at Chair of Political Science, Quantitative Methods in the Social Sciences**, University of Mannheim (Thomas Gschwend)

09/2021-12/2021 **Student Research Assistant at Project Data Science and Big Data at Schools Prodabi**, Paderborn University (Rolf Biehler)

# Selected Work Experience (Industry)

07/2018-02/2020 **Working Student Data Science/Finance**, SAP SE in Walldorf, Germany

02/2018-07/2018 **Internship Custom Solution Development**, Capgemini in Stuttgart, Germany

09/2012-07/2015 **Traineeship Software Developer**, KUMAVISION AG in Tiengen, Germany